bridges between past, present, and future. To a large extent, the brain is a mechanism for collecting memories of the past in order to use them continually to predict the future. This happens across a wide spectrum of time scales, from the very short to the very long. If someone throws something at us to catch, our hand moves skillfully to the place where the object will be in a few instants: the brain, using past impressions, has very rapidly calculated the future position of the object that is flying toward us. Further along the scale, we plant seed so that corn will grow. Or invest in scientific research so that tomorrow it might result in knowledge and new technology. The possibility of predicting something in the future obviously improves our chances of survival and, consequently, evolution has selected the neural structures that allow it. We are the result of this selection. This being between past and future events is central to our mental structure. This, for us, is the "flow" of time.

There are elementary structures in the wiring of our nervous system that immediately register movement: an object that appears in one place and then immediately afterward in another does not generate two distinct signals that travel separately toward the brain, but a single signal correlated with the fact that we are looking at something that is moving. In other words, what we perceive is not the present, which in any case makes no sense for a system that functions on a scale of finite time, but rather something that happens and extends in time. It is in our brains that an extension in time becomes condensed into a perception of duration.

This intuition is an ancient one. Saint Augustine's ruminations on it have remained famous.

In Book XI of the *Confessions*, Augustine asks himself about the nature of time and, despite being interrupted by exclamations in the style of an evangelical preacher that I find quite tiresome, he presents a lucid analysis of our capacity for perceiving time. He observes that we are always in the present, because the past is past and therefore does not exist, and the future has yet to arrive, so it does not exist either. And he asks himself how we can be aware of duration—or even be capable of evaluating it—if we are always only in a present that is, by definition, instantaneous. How can we come to know so clearly about the past, about time, if we are always in the present? Here and now, there is no past and no future. Where are they? Augustine concludes that they are within us:

> It is within my mind, then, that I measure time. I must not allow my mind to insist that time is something objective. When I measure time, I am measuring something in the present of my mind. Either this is time, or I have no idea what time is.